Module 2 Assignment

VSD 7nm FinFET Workshop

For my username, **<u>mansi</u>**, the ASCII sum = 109 + 97 + 110 + 115 + 105 = 536

Therefore voltage = 0.536 mv

inverter_vtc.spice

```
** sch_path: /home/hprcse/Finfet/inverter_vtc.sch
**.subckt inverter_vtc
Xpfet1 nfet_out nfet_in vdd vdd asap_7nm_pfet l=7e-009 nfin=28
Xnfet1 nfet_out nfet_in GND GND asap_7nm_nfet l=7e-009 nfin=21
V1 nfet_in GND pulse(0 0.7 20p 10p 10p 20p 500p 1)
V2 vdd GND 0.536
**** begin user architecture code


.dc v1 0 0.7 1m
```

1) Simulate the inverter using the provided SPICE deck

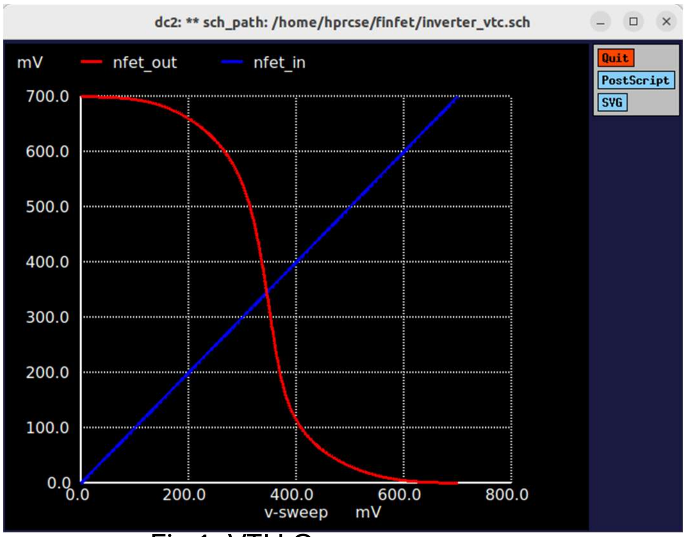


Fig 1. VTH Curve

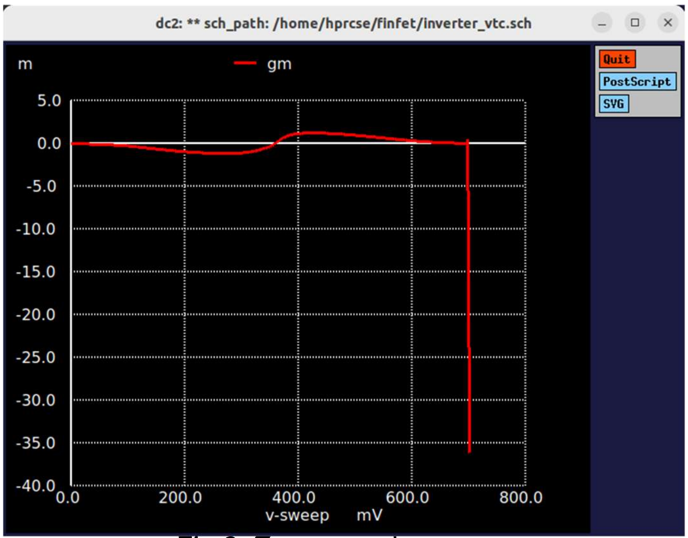


Fig 2. Transconductance

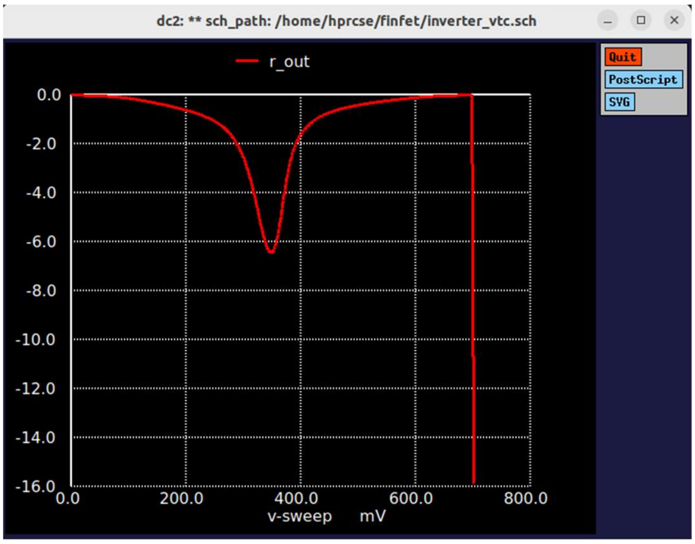


Fig 3. Output Resistance

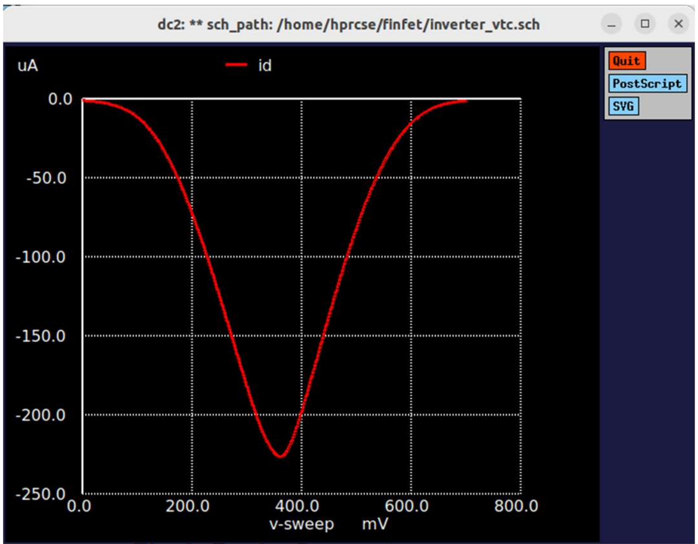
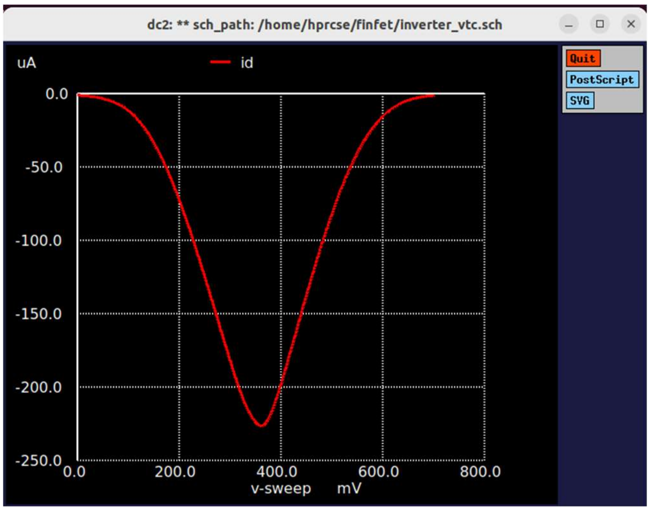


Fig 4. Drain Current

2) Modify parameters (W, L) to observe performance variation
   i) NMOS W/L = 2 ; PMOS W/L = 2
   ii) NMOS W/L = 2 ; PMOS W/L = 3
   iii) NMOS W/L = 3 ; PMOS W/L = 2

3) Extract the following metrics from simulation results
   i) ID = 225uA



   ii) Gain, Noise Margin, Transconductance

```
Using SPARSE 1.3 as Direct Linear Solver
 Reference value :  6.67000e-01
No. of Data Rows : 701
v_th                 =  3.447862e-01
max_gain             =  6.428448e+00 at=  3.500000e-01
vil                  =  3.488156e-01
voh                  =  3.190767e-01
vil                  =  3.510961e-01
voh                  =  3.044187e-01
Warning from checkvalid: vector vih is not available or has zero length.
Error: RHS "voh - vih" invalid
Warning from checkvalid: vector vol is not available or has zero length.
Error: RHS "vil - vol" invalid
Warning from checkvalid: vector vih is not available or has zero length.
gm_max               =  1.235805e-03 at=  4.250000e-01
Doing analysis at TEMP = 27.000000 and TNOM = 27.000000
```

   iii) Power Consumption and Propagation Delay

```
Initial Transient Solution
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Node                                   Voltage
----                                   -------
nfet_out                              0.699647
nfet_in                                      0
vdd                                        0.7
in                                       0.536
vuniq#branch                                 0
v2#branch                          -8.07135e-07
v1#branch                           7.22294e-12

 Reference value :  5.01000e-11
No. of Data Rows : 120
tpr                  =  2.500000e-11
tpf                  =  2.560432e-11
id_pwr               =  -1.69406e-15 from=  2.00000e-11 to=  6.00000e-11
tpr = 2.500000e-11
tpf = 2.560432e-11
tp = 2.530216e-11
id_pwr = -1.69406e-15
pwr = -1.18584e-15
power = 2.964598e-05
Doing analysis at TEMP = 27.000000 and TNOM = 27.000000
```

iv) Frequency

```
Using SPARSE 1.3 as Direct Linear Solver

Initial Transient Solution
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Node                                    Voltage
----                                    -------
nfet_out                               0.699647
nfet_in                                       0
vdd                                         0.7
in                                        0.536
vuniq#branch                                  0
v2#branch                          -8.07135e-07
v1#branch                           7.22294e-12

 Reference value :  7.50000e-11
No. of Data Rows : 71
tr                    =  2.100000e-11
tf                    =  2.351715e-11
t_delay = 4.451715e-11
f = 2.246325e+10
```

4) Characterization Table

S. No	W/L PMOS	W/L NMOS	Vth (V)	Id (A)	P (W)	tpd (ps)	Av	f (Hz)
1	2	2	3.44786e-01	225u	2.9646e-05	2.53e-11	6.428	2.246e+10
2	2	3	3.2783e-01	275u	4.0458e-05	2.52e-11	6.573	2.263e+10
3	3	2	3.6952e-01	272u	4.1032e-05	2.56e-11	6.395	2.1968e+10

Observations:

- Increasing the Pfet width shifts the VTC curve to the right.
- Increasing the Nfet width shifts the VTC curve to the left.
- Increasing the width of either the Nfet or Pfet results in a higher drain current.
- Increasing the Nfet size leads to an increase in gain.
- Increasing the Pfet size causes the gain to decrease.
- Power increases whenever the size of either device is increased.
- Drain current goes up as the drive strength increases.